

Cambridge Lower Secondary Checkpoint

CANDIDATE
NAME

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CENTRE
NUMBER

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CANDIDATE
NUMBER

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MATHEMATICS

0862/02

Paper 2

April 2025

1 hour

You must answer on the question paper.

You will need: Geometrical instruments
Tracing paper (optional)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You should show all your working in the booklet.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages. Any blank pages are indicated.

- 1 On Wednesday Carlos takes 16 minutes to cycle home from work.
On Thursday he does the same journey at half the speed.

Work out how many minutes his journey takes on Thursday.

..... mins [1]

- 2 Calculate $\sqrt{7} \times \sqrt{28}$

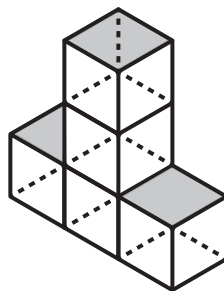
..... [1]

- 3 Jamila thinks of a number n .
She adds 7 to the number.
Then she squares the result.

Write down an algebraic expression, in terms of n , for the number Jamila finishes with.

..... [1]

- 4 Here is a prism made from five identical cubes.



Draw a ring around the number of planes of symmetry this prism has.

0

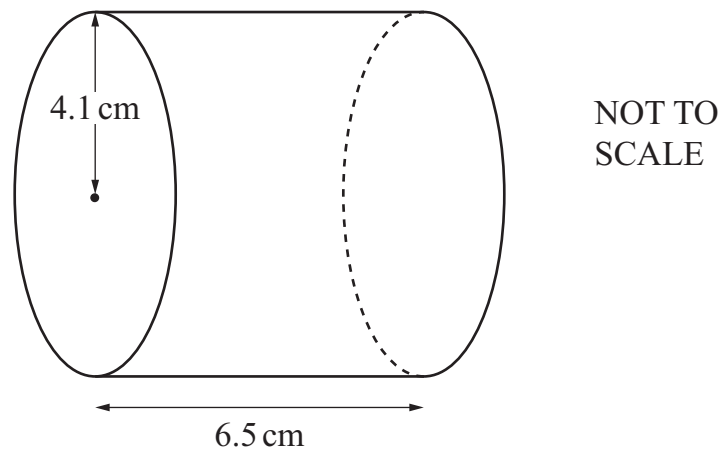
1

2

3

[1]

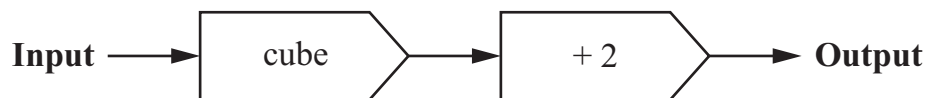
- 5 The diagram shows a cylinder with a radius of 4.1 cm and a length of 6.5 cm.



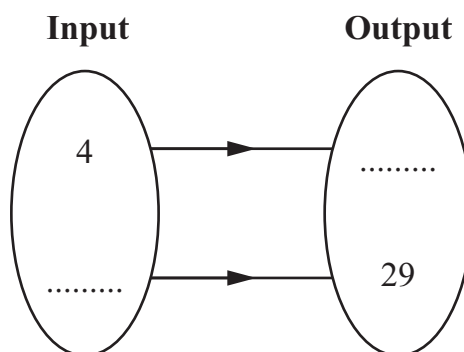
Calculate the volume of the cylinder.

..... cm³ [2]

- 6 Here is a function machine.

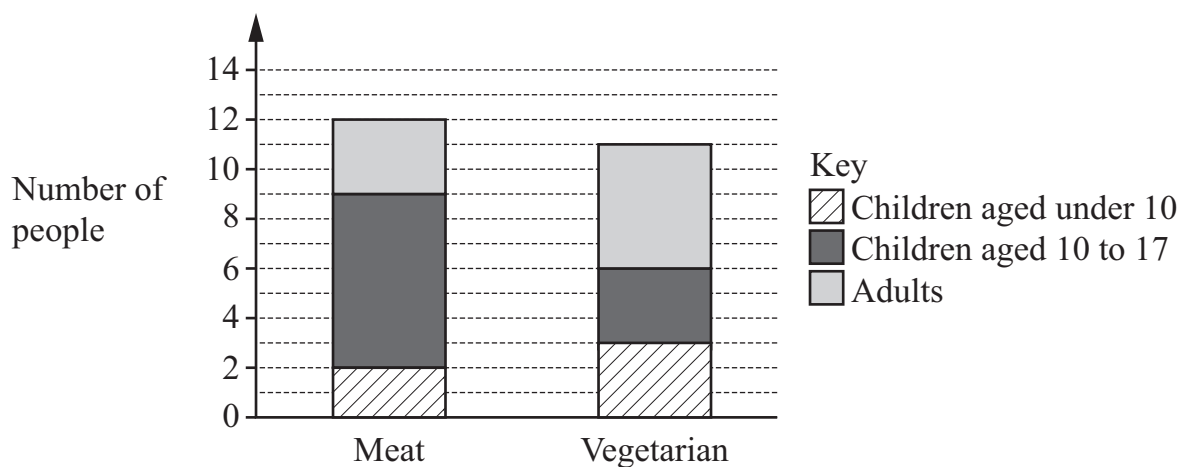


Complete the mapping diagram for this function.



[2]

- 7 The compound bar chart shows information about the favourite types of pizza chosen by a group of children and adults.
Each person in the group chooses either meat or vegetarian.



Mike draws this table to show the data from the compound bar chart.

	Meat	Vegetarian
Children aged under 10	2	3
Children aged 10 to 17	9	6
Adults	12	11

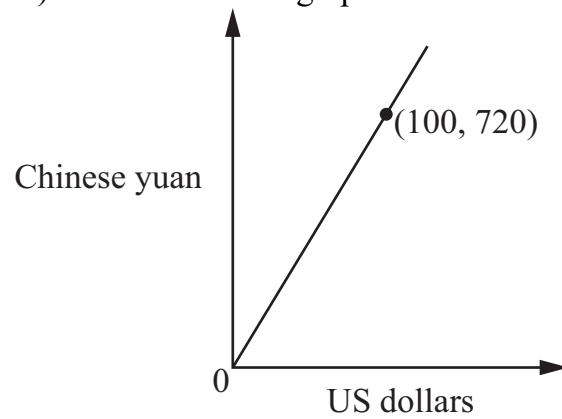
Mike has made some mistakes in his table.

Complete the table correctly.

	Meat	Vegetarian
Children aged under 10	2	3
Children aged 10 to 17		
Adults		

[2]

- 8 Here is a graph showing the exchange rate between US dollars and Chinese yuan.
The coordinate (100, 720) is marked on the graph.



Calculate the exchange rate.
Give your answer in yuan per dollar.

..... yuan per dollar [1]

- 9 A line has the equation $4y = 5x + 16$

Find the coordinates of the y -intercept of this line.

(..... ,) [1]

- 10 Seven students each solve two maths questions.

This back-to-back stem-and-leaf diagram shows the time taken for each student to solve question 2

Question 1						Question 2				
					3	6	6			
					4	1	3			
					5	0	2	9		

Key: 4 | 3 | 6 represents 34 seconds to solve question 1
and 36 seconds to solve question 2

Here are the times, in seconds, it takes the seven students to solve question 1

34 50 41 51 47 46 41

- (a) Complete the back-to-back stem-and-leaf diagram.

[2]

- (b) Complete the table.

	Question 1	Question 2
Median	46	
Mode	41	
Range	17	23

[2]

(c) Eva says,

‘The times students take to solve question 1 are less consistent than the times students take to solve question 2’

Complete the sentence using one word from each box to explain how you know Eva is **not** correct.

median	mode	range
--------	------	-------

higher	lower
--------	-------

The for question 1 is
than for question 2

[1]

11 A regular polygon has 10 sides.

Calculate one interior angle of the polygon.

..... ° [2]

- 12 (a)** A number, n , rounded correct to the nearest 10 is 380

Complete the inequality to show the bounds for n .

$$\text{.....} \leq n < 385$$

[1]

- (b)** The mass of a baby is 3.62 kg correct to 2 decimal places.

Find the upper bound of the mass of the baby.

$$\text{..... kg} \quad [1]$$

- 13** In a sports club the ratio of adults to children is 11 : 6
There are 750 **more** adults than children in the sports club.

Calculate the number of children in the sports club.

$$\text{.....} \quad [2]$$

- 14 M is the midpoint of the line AB .
 M is the point $(5, 8)$.

Pierre writes some statements about possible coordinates of A and B .

Tick (✓) to show if each of these statements could be correct or is **not** correct.

	Could be correct	Not correct
A is the point $(-2, 8)$ and B is the point $(12, 8)$.	☐	☐
A is the point $(-2, 4)$ and B is the point $(8, 12)$.	☐	☐

[1]

- 15 A painting costs \$17 400
 The painting increases in value by 4% each year.
 The calculation to find the value of this painting after 5 years is $17\,400 \times x^5$
 Find the value of x .

$x =$ [1]

- 16 In this question x is always positive.

For each function, tick (✓) to show if the statement is true or false.

Function	Statement	True	False
$y = x + 3$	As x increases y increases		
$v = 5x$	As x increases v increases		
$w = 50 - x$	As x increases w increases		
$z = \frac{x}{5}$	As x increases z increases		

[2]

- 17** Hassan picks a letter at random from a book.

The letters in the book are vowels (A, E, I, O, U) or consonants.

The table shows the probabilities of Hassan picking some of the letters.

	Vowels					Consonants
	A	E	I	O	U	
Probability	0.06		0.07	0.1	0.04	

There are twice as many of the letter **E** in the book as there are of the letter **I**.

Find the probability of Hassan picking a letter from the book that is a **consonant**.

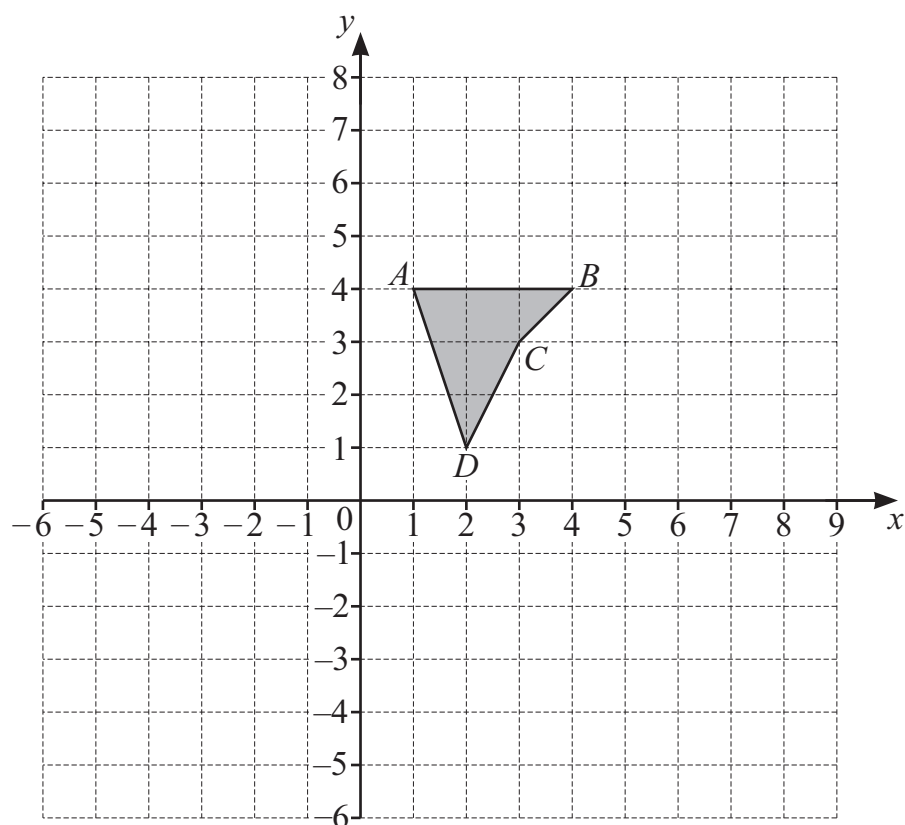
..... [3]

- 18** Solve the inequality.

$$11x > 6x + 32$$

..... [2]

19 The diagram shows shape $ABCD$ on a grid.



$ABCD$ is enlarged by scale factor 2, centre $(5, 7)$ to $A'B'C'D'$.

Complete the table to show the positions of vertices A' , B' , C' and D' .
One has been done for you.

	Positive y -coordinate	Negative y -coordinate
Positive x -coordinate	B'	
Negative x -coordinate		

[2]

- 20 The map shows the positions of a shop, S , and a pond, P .
The scale of the map is 1 centimetre represents 50 metres.



Scale: 1 cm to 50 m



A restaurant, R , is on a bearing of 135° from the shop.
The restaurant is 175 metres from the pond.

Mark the **two** possible positions of the restaurant on the map.

[3]

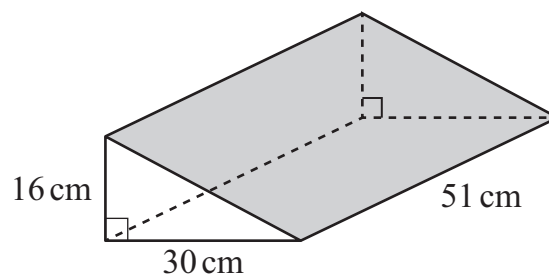
21 Expand and simplify.

$$(x + 3)^2$$

..... [2]

22 The diagram shows a triangular prism.

The cross-section of the prism is a right-angled triangle.



NOT TO
SCALE

Calculate the area of the **shaded** face of the prism.

..... cm^2 [2]

23 Solve.

$$\frac{27}{4x-3} = 5$$

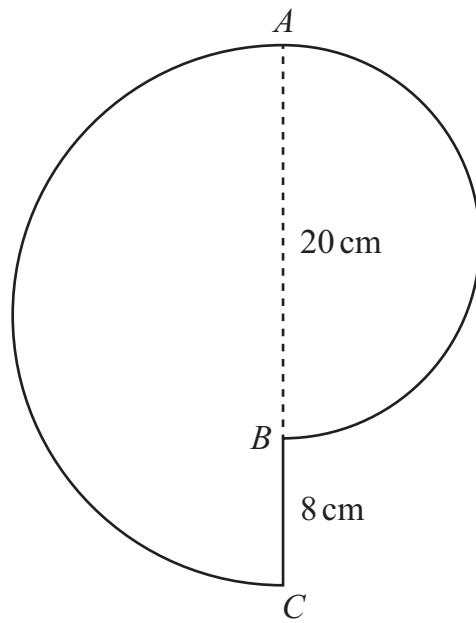
$$x = \dots\dots\dots [3]$$

24 Anastasia thinks of two numbers t and y .
 $(t - y)$ is a third of $(t + y)$.

Write t in terms of y .

$$t = \dots\dots\dots [2]$$

25 Here is a shape made from two semicircles.



NOT TO
SCALE

The larger semicircle has a diameter AC .

The smaller semicircle has a diameter AB .

ABC is a straight line with $AB = 20$ cm and $BC = 8$ cm.

Calculate the **perimeter** of the shape.

Give your answer correct to 1 decimal place.

..... cm [4]

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